

## ARCHITECT REVIEW

# SALDEV-1413 | Stepped-up Pricing ARR Flows

Flywire Partial sandbox | Read-only implementation and promotion-readiness assessment

|             |                                                                                 |
|-------------|---------------------------------------------------------------------------------|
| Environment | FlywirePartial   Org 00DhG0000000jOX   Sandbox                                  |
| Review date | 26 August 2026                                                                  |
| Evidence    | Active-state read-back   retrieved metadata   Apex test   SOQL collision checks |
| Decision    | Changes Requested - not ready for promotion                                     |
| Boundary    | No code edits, deployment, activation, data changes, or Production changes      |

### ARCHITECT DECISION

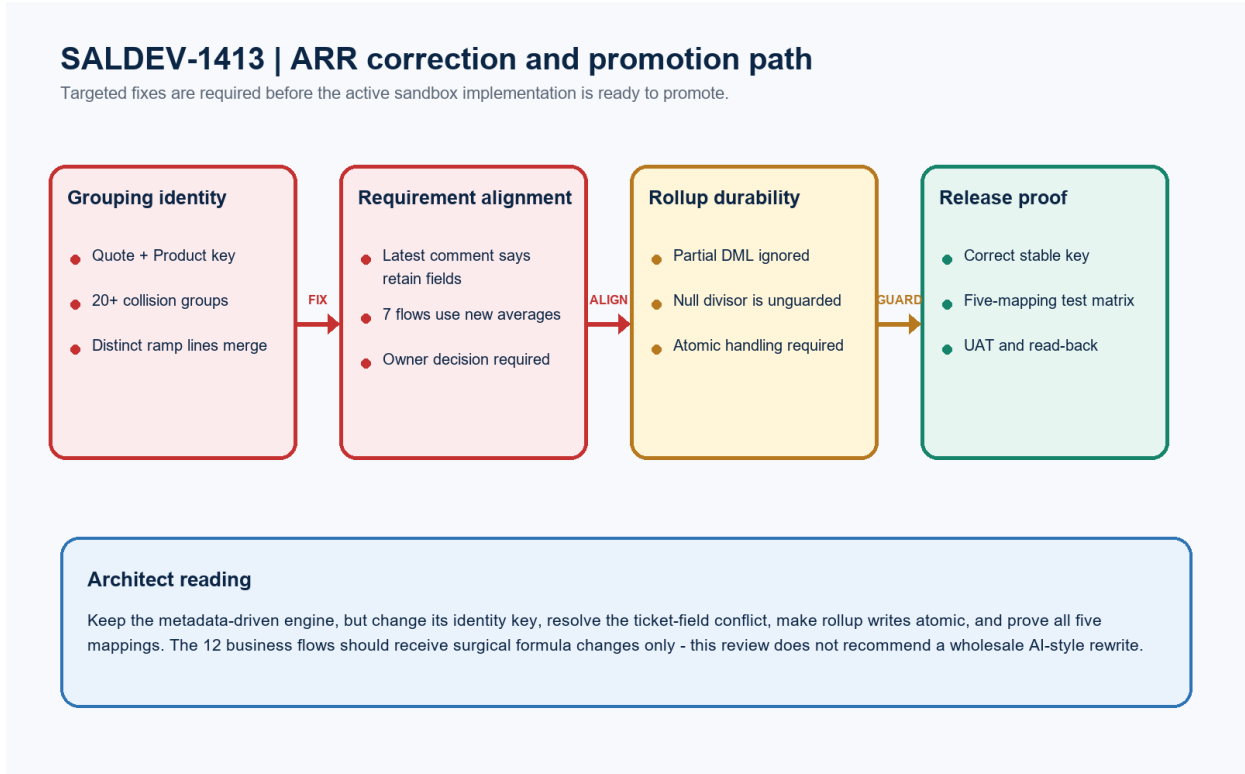
**Do not promote yet. The implementation is bulk-aware and metadata-driven, but its aggregation key merges distinct lines, seven active flows conflict with the latest story clarification, and rollup failures can leave unsafe prorate divisors.**

## Acceptance outcome

| # | Scope                   | Result  | Evidence                                                                                                       |
|---|-------------------------|---------|----------------------------------------------------------------------------------------------------------------|
| 1 | Aggregation correctness | Fail    | Apex groups only by Quote + Product; sandbox evidence found 20+ groups with multiple distinct line identities. |
| 2 | Requirement alignment   | Fail    | The final story comment says retain Quoted Usage fields, but two mappings and seven active flows average them. |
| 3 | Runtime durability      | Fail    | Partial SaveResults are ignored and three flows divide directly by Total_Prorate_Multiplier__c.                |
| 4 | Active component scope  | Pass    | All 12 named target flows are active/latest; the active B2B XB API name is the _1 definition.                  |
| 5 | Automated proof         | Partial | Focused Apex tests passed, but handler coverage is 72% and four mappings plus Flow behavior are untested.      |

# Correction and control path

Use a targeted correction: preserve the metadata-driven rollup engine, group by a stable ramp-line identity, align the seven affected flows with the final product-owner decision, make rollup writes atomic, and prove the result with focused Apex and business UAT.



## Current SALDEV-1413 implementation evidence

### Apex and configuration evidence

| Evidence area       | Current implementation   | Acceptance | Architect reading                                |
|---------------------|--------------------------|------------|--------------------------------------------------|
| Group identity      | Quote + Product          | Fail       | Distinct ramp lines can merge.                   |
| Requirement mapping | 5 CMDT mappings          | Fail       | Two usage mappings contradict the final comment. |
| DML handling        | update(..., false)       | Fail       | Partial SaveResults are unchecked.               |
| Prorate safety      | Direct division          | Fail       | Three flows lack divisor guards.                 |
| Audit narrative     | Healthcare text mismatch | Fail       | Formula and audit text use different fields.     |
| Bulk structure      | 2 queries + 1 DML        | Pass       | The bulk execution pattern is sound.             |

### Flow and test evidence

| Evidence area      | Observed state      | Acceptance | Architect reading                           |
|--------------------|---------------------|------------|---------------------------------------------|
| Usage-field flows  | 7 active flows      | Fail       | Seven flows change fields marked to retain. |
| Prorate flows      | 3 active flows      | Fail       | Three flows divide without a guard.         |
| Collision evidence | 20+ groups          | Fail       | 395 lines / 394 identities in one group.    |
| Test coverage      | 5 methods; 72%      | Fail       | Only the rate mapping is asserted.          |
| Definition state   | 12/12 active/latest | Pass       | All target definitions match.               |

#### PROMOTION BLOCKER

Do not promote until duplicate-line grouping, usage-field alignment, divisor safety, and all five mappings are proven.

## Implementation and release boundary

| Component type      | Focused scope                                 | Required action                                                                                   |
|---------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------|
| Apex rollup service | QuoteLineTriggerHandler.calculateAveragePrice | Group by Opportunity_Line_Id__c or an approved composite identity.                                |
| Quote Line trigger  | QuoteLineTrigger after-save path              | Preserve bulk execution; make recursion reset and error handling deterministic.                   |
| Custom metadata     | 5 Quote Line rollup mappings                  | Validate fields and operations; retain/remove usage mappings after owner decision.                |
| Usage ARR flows     | 7 active flows                                | Apply only the approved usage-field formula decision; preserve unrelated logic.                   |
| Prorate ARR flows   | Edu Payables; Fixed Price; Travel Payables    | Guard null/zero totals and require atomic rollup completion.                                      |
| Automated tests     | Apex plus targeted Flow/UAT scenarios         | Cover five mappings, duplicate identities, bulk quotes, delete/update, null/zero, and regression. |

## Release sequence

| Phase                   | Required evidence                                                                                                                                                          |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pre-deployment          | Obtain written owner confirmation on Quoted Usage fields; approve the stable line identity; retrieve the current target-org baseline.                                      |
| Sandbox correction      | Make a focused Apex/mapping/formula change only; preserve unrelated Flow behavior and active-state boundaries.                                                             |
| Validation and UAT      | Run the expanded Apex matrix, focused Flow tests, check-only validation, stepped/non-stepped business scenarios, and capture job IDs and record-level expected ARR.        |
| Promotion and read-back | Promote only after approval; verify active versions, Apex coverage, rollup fields, ARR outputs, and logs. Roll back the approved package if a critical regression appears. |

### PRODUCTION GATE

**This review covered FlywirePartial only. Production was not inventoried or changed. Re-retrieve current Production metadata and run focused check-only validation before any deployment decision.**

## UAT and approval gates

| Test                   | Pass evidence                                                                                                                              |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Stepped grouping       | Two occurrences of the same product retain separate identities while each ramp sequence receives its own average and total proration.      |
| Non-stepped regression | A non-stepped line produces the same ARR as the current approved baseline with no average-field side effect.                               |
| Usage-field decision   | Percent and amount scenarios match the written product-owner decision, including the existing divide-by-100 behavior.                      |
| Prorate products       | Edu Payables, Fixed Price, and Travel Payables handle populated, null, and zero total-prorate conditions without blank ARR or Flow faults. |
| Bulk and lifecycle     | Multi-quote insert, update, delete, start-date shift, and duplicate-product tests pass with expected rollup placement.                     |

## Required approval gates

| # | Required action                 | Owner                     | Evidence needed                                                                               |
|---|---------------------------------|---------------------------|-----------------------------------------------------------------------------------------------|
| 1 | Confirm usage-field requirement | Product Owner / Architect | Written decision resolving the final comment versus earlier design text.                      |
| 2 | Approve rollup identity         | CPQ Architect             | Named stable key and examples for repeated products, bundles, and ramp groups.                |
| 3 | Complete focused correction     | Development Lead          | Reviewed Apex, CMDT, and Flow diff with no unrelated rewrite.                                 |
| 4 | Validate sandbox package        | Release Manager / QA      | Expanded test results, coverage, check-only job ID, UAT records, and expected values.         |
| 5 | Promote and read back           | DevOps / Business Owner   | Deployment receipt, active-version checks, field-level read-back, ARR results, and sign-offs. |

## Ready-to-paste architect response

### STORY COMMENT

Architect review completed for SALDEV-1413 in FlywirePartial. Changes Requested. All 12 target flows are active/latest and focused Apex tests passed, but Quote + Product grouping can merge distinct line identities. The final story comment says retain Quoted Usage Percent and Amount; two mappings and seven active flows average them. Partial rollup failures are ignored, and three flows divide by Total\_Prorate\_Multiplier\_\_c without a null/zero guard. Confirm the usage-field requirement, approve a stable line key, make writes atomic, protect the divisor, and run the five-mapping stepped/non-stepped matrix before promotion. Retrieval 09ShG000006FuUgUAK; Apex 707hG00000LJDjIQA. No code, metadata, activation, deployment, data, or Production changes were made.